

IoTsmart RP2350A v.2.02

IoTsmart RP2350A: Dual-Architecture Automation Controller

Built for control, not just connectivity — our latest module is engineered around **Raspberry Pi's RP2350A** microcontroller. This powerful dual-core, dual-architecture MCU isn't just a component; it's the core of a solution engineered to deliver precise, deterministic control for demanding automation tasks.



Discover the power of the **IoTsmart RP2350A** module, engineered around the advanced **Raspberry Pi's RP2350A** microcontroller, packing a dual-core and dual-architecture design: dual-core **Arm Cortex-M33** processor and dual-core **Hazard 3 RISC-V** processor, flexible clock running up to 150 MHz, 520KB of SRAM, and 4MB of on-board flash memory.

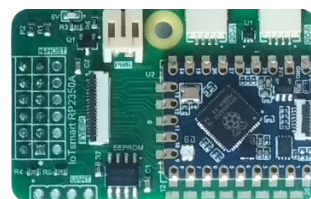
Key enhancements in version 2.02 compared to **1.xx** versions:

- **Optimized Form Factor:** The module now measures 47x30 mm, providing better mechanical compatibility and a more seamless fit when docked with **IoTextra** mezzanines.
- **EYESPI Interface:** A new onboard **EYESPI** connector adds plug-and-play support for popular LCDs and other displays using a standardized 18-pin FPC cable.
- **Dual Qwiic® Connectors:** The addition of a second **Qwiic** connector allows for easy daisy-chaining of multiple **I²C** devices and sensors without the need for additional hubs.

Whether you're innovating in IoT, crafting smart home solutions, or deploying complex industrial controls, the **IoTsmart RP2350A** connects seamlessly with **IoTextra** series expansion boards (mezzanines) via a 12-pin HOST connector. Essentially, this module upgrades standard mezzanines into intelligent, programmable smart modules.

The module offers two orientations for the **HOST** connector:

- **H-HOST** (horizontal connector), mounted on the **top-side** of the module, designed for **vertical** mezzanine installation
- **V-HOST** (vertical connector), mounted on the **bottom-side** of the module, designed for **horizontal** mezzanine installation



IoTsmart RP2350A
module (top-side)



IoTsmart RP2350A
module (bottom-side)

The module is equipped with two **Qwiic®** connectors for **I²C** communication and supports optional **EYESPI** (for connecting displays). **UART** connector is supported but must be installed by the user.

An onboard EEPROM (8 Kbit or 16 Kbit) stores configuration and user data and is accessible via **I²C** at addresses 0x54–0x57 by default.

The module incorporates **Waveshare RP2350-Tiny**, which provides:

- Dual-core Arm Cortex-M33 processor and dual-core Hazard 3 RISC-V processor,
- Flexible clock running up to 150 MHz
- 520KB SRAM, 4MB flash memory
- USB Type-C interface via the **Tiny Adapter Board**
- FPC 8-pin connector

The module is powered from a +5VDC input supply.

The **IoTsmart RP2350A** module dimensions are 47 x 30 mm.

COMMON APPLICATIONS

 **IIoT — Modbus/RS485 Field I/O & Precision Actuator Control.** The module operates as a Modbus RTU slave over UART, exposing relays, pressure/temperature sensors, and limit switches from a control cabinet to a master controller. The dual-core, dual-architecture design provides deterministic timing for precise actuator switching without network-stack latency. Master role is handled by a PLC, SCADA gateway, or **Node-RED** host over **Modbus RTU (UART)**.

 **Smart Home — Deterministic I/O Expansion.** The module performs exact, real-time control of physical outputs — relay timing, dimmer phase-angle calculations — without an onboard network stack. It communicates over **Modbus RTU (UART)** with any low-cost external WiFi module (e.g. **ESP32-C3**) bridging to **Home Assistant**, which handles connectivity and platform integration.

 **Local HMI & Display Control.** With the optional EYESPI connector, the module drives a local TFT display directly — sensor readings, status indicators, a settings menu — rendered with **LVGL**, entirely without network dependency.

QUICK START

The **IoTsmart RP2350A** module is primarily designed for use with **IoTextra** series mezzanine. This combination allows the **IoTsmart RP2350A** module and a mezzanine to easily function as a standalone smart device.

The photos below illustrate the **IoTsmart RP2350A** module paired with various mezzanines:



IoTsmart RP2350A
with **IoTextra Input**



IoTsmart RP2350A
with **IoTextra SSR Small**

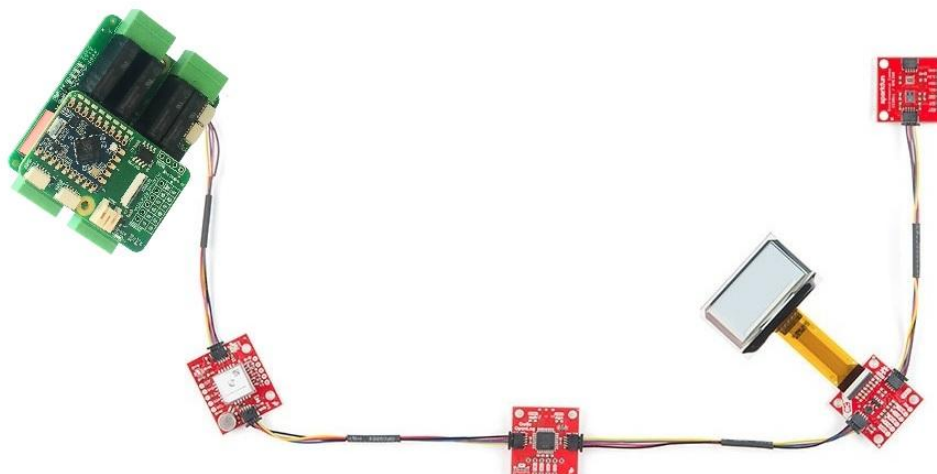


IoTsmart RP2350A
with **IoTextra Octal2**



IoTsmart RP2350A
with **IoTextra Analog**

The **IoTsmart RP2350A** also readily connects to numerous [Qwiic®](#) compatible sensors, peripherals and modules via the **I²C** connector:



WAVESHARE RP2350-TINY

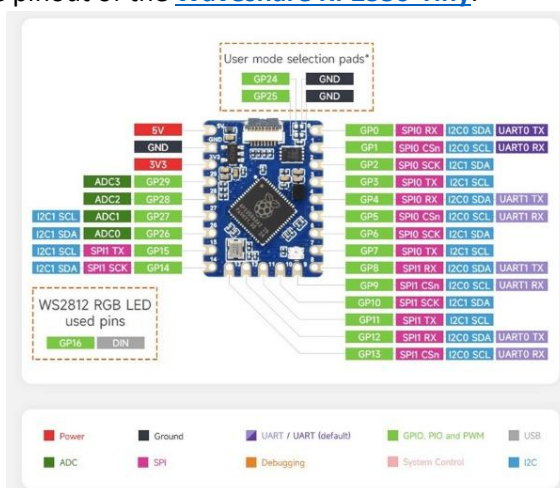
The **IoTsmart RP2350A** module utilizes the [Waveshare RP2350-Tiny](#), which adopts unique dual-core and dual-architecture design: dual-core Arm Cortex-M33 processor and dual-core Hazard 3 RISC-V processor, flexible clock running up to 150 MHz.

The **Waveshare RP2350-Tiny** features 520KB of SRAM, and 4MB of onboard flash memory. It also includes an on-board FPC 8-pin connector, which adapts the USB Type-C port via the **Tiny Adapter Board**.

The following photo displays the **Tiny Adapter Board** with a cable; however, these are not included with the **IoTsmart RP2350A** and must be purchased separately:



For your information, here is the pinout of the [Waveshare RP2350-Tiny](#):

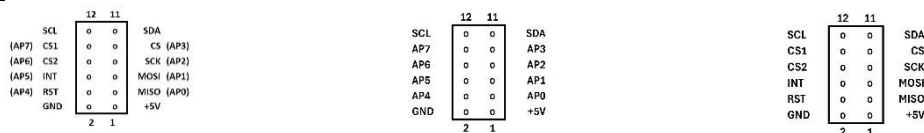


CONNECTORS

The module is equipped with the following connectors:

- **H-HOST** - horizontal (right-angle) connector installed on the **top-side**, designed for **vertical** mezzanine installation, **V-HOST** - vertical connector installed on the **bottom-side**, designed for **horizontal** mezzanine installation. Information regarding the type of the installed **HOST** connector is provided on the **bottom-side** of the module
- **EYESPI** connector is intended for communication with displays (indicators) using an 18-pin FFC cable.
- Two **Qwiic**® connectors for connecting external sensors and devices via the **I²C** bus.
- An optional **UART** connector (not pre-installed)
- A **PWR** connector (+5VDC input)

HOST connector. Pinout of the **HOST** connectors:



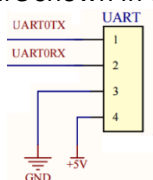
HOST on the **IoTsmart** module

HOST-P12 on the mezzanine

HOST-S on the mezzanine

The **HOST** connector of the **IoTsmart RP2350A** module is inserted into the **HOST-12** or **HOST-S** connector on the mezzanine. Therefore, the pinout for the **HOST-P12** and **HOST-S** connectors on the mezzanines is also shown in the figure to compare the connector signals.

UART connector. The contacts of this connector are shown in the figure:



EYESPI Connector. The module's top side has an **EYESPI** connector. It is intended for communication with displays (indicators) using an 18-pin FFC cable. Communication with indicators is achieved due to compatibility with the 18-pin **EYESPI Cable** used by **Adafruit**. The standardization of this cable allows the following displays to be connected:

- [2.0" 320x240 Color IPS TFT Display with microSD Card Breakout](#)
- [2.2" 18-bit color TFT LCD display with microSD card breakout](#)
- [1.8" Color TFT LCD display with MicroSD Card Breakout](#)
- [OLED Breakout Board - 16-bit Color 1.5" w/microSD holder](#)
- [OLED Breakout Board - 16-bit Color 1.27" w/microSD holder](#)
- [Adafruit 2.7" Tri-Color eInk](#)
- [Adafruit 1.54" Monochrome eInk](#)

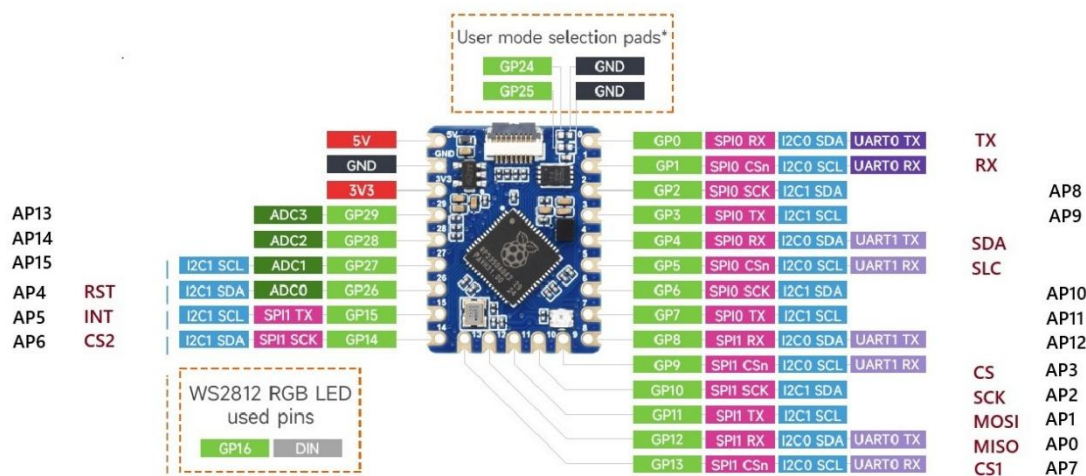
Waveshare 2" Capacitive Touch Display Module can also be connected using an 18-pin FFC cable.

Here is the correspondence between the module's signals and the signals in the **EYESPI** connector:

Here is the correspondence between the module's signals and the signals in the **EYESPI** connector:

EYESPI		Signals on the IoTsmart RP2350A module schematic
pin #	NAME	
1	VCC	+3V3
2	BACKLITE	AP9
3	GND	GND
4	SCK	AP2
5	MOSI	AP1
6	MISO	AP0
7	DC	RX
8	RESET	AP4
9	DISP_CS	TX
10	SD_CS	AP3
11	MEM_CS	AP6
12	TC_CS	AP8
13	SCL	SCL
14	SDA	SDA
15	INT	AP5
16	GP1	AP10
17	GP2	AP11
18	BUSY/TE	AP7

The **HOST** and **EYESPI** connector signals correspond to the **Waveshare RP2350-Tiny** signals as follows:



EEPROM

To store configuration and other user information, the **IoTsmart RP2350A** module includes an onboard **EEPROM** (8 Kbit or 16 Kbit). This **EEPROM** is accessible via the I²C bus and is visible at addresses 0x54-0x57 by default.

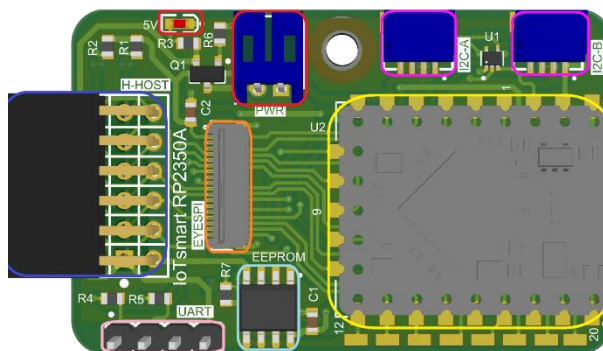
POWER SUPPLY

The module operates from a 5VDC power input supply. The power supply can be connected via a 2-pin **PWR** connector (JST S2B-PH-K-S, 2.00 mm pitch) or through the **Tiny Adapter Board**.

The typical power consumption of the module with the **Waveshare RP2350-Tiny** is approximately 25 mA (measured without additional connected peripherals or expansion boards). The maximum current consumption, including power provided to a mezzanine, should not exceed 1000 mA.

LAYOUT

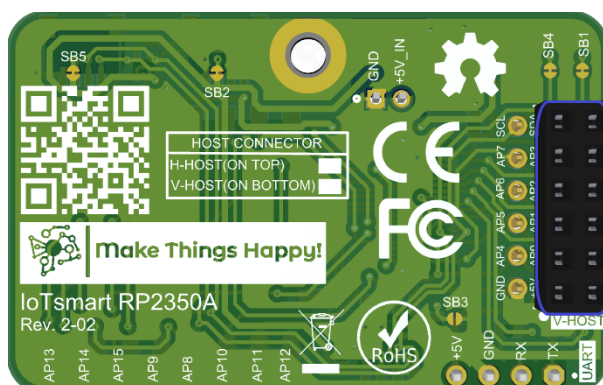
Below is the layout of the elements on the **top-side** of the **IoTsmart RP2350A** module when using the **HOST-H** connector:



In this picture:

- Power-related elements (external power connector, named **PWR**, and **LED**) are highlighted in **red**
- The soldered **Waveshare RP2350-Tiny** is highlighted in **yellow**
- The **HOST-H** connector is highlighted in **blue**
- The **EYESPI** connector is highlighted in **orange**
- **UART** connector not installed during production is highlighted in **pink**
- The **Qwiic**® connectors for connecting peripherals via the **I²C** bus are highlighted in **purple**
- **EEPROM** is highlighted in **light blue**

Below is the layout of the **IoTsmart RP2350A** module **bottom-side** when using the **HOST-V** connector (highlighted in **blue**):



JUMPERS

Jumpers are located on the underside of the module:

- **SB1, SB4** – Connect pull-up resistors to **I²C** SCL and SDA (enabled by default)
- **SB2, SB5** –Disable +3.3V power to devices when open (default: closed)
- **SB3** – Sets EEPROM **I²C** address:

b7	b6	b5	b4	b3	b2	b1	b0
1	0	1	0	1	x	x	R/W

CONFIGURATION TABLES

The **bottom-side** of the module provides information about the type of **HOST** connector installed:

- **H-HOST** - horizontal (right-angle) connector installed on the **top-side**, designed for **vertical** mezzanine installation.
- **V-HOST** - vertical connector installed on the **bottom-side**, designed for **horizontal** mezzanine installation.



COMPATIBILITY WITH MEZZANINES

The **IoTsmart RP2350A** is compatible with all **IoTextra** mezzanines, including

Digital IoTextra	Analog IoTextra	Combo IoTextra
• IoTextra Input • IoTextra Relay • IoTextra Relay2 • IoTextra SSR Small • IoTextra MOSFET2 • IoTextra Octal • IoTextra Octal2 • IoTextra Octal3 • IoTextra Octal4	• IoTextra Analog • IoTextra Analog2 • IoTextra Analog3	• IoTextra Combo • IoTextra Combo2 • IoTextra Omni

ACCESSORIES

Recommended accessories:

- Two-pin cable with a JST connector (for Power Supply)
- **Tiny Adapter Board** with USB Type-C
- **Qwiic® I²C** cable (connector on both ends)
- 18-pin FPC standard **EYESPI** cable for connecting a display (indicator) or inter-module connection, 100mm or 200mm long